RESUME OF WEICHENG YE

victoryeofphysics@gmail.com
(+1)778-847-2018

POSITION

University of British Columbia
Postdoc *Sep 2023 -*

EDUCATION

University of Waterloo/Perimeter Institute of Theoretical Physics
Advisor: Prof. Chong Wang and Prof. Timothy Hsieh
Doctor of Philosophy in Physics *Aug 2019- Oct 2023*

Stony Brook University
Master of Arts in Physics *Aug 2016 - Aug 2018*

Peking University
Advisor: Prof. Bin Chen
Bachelor of Science in Physics *Sep 2012 - Jul 2016*

INTERESTS

Theoretical Condensed Matter Physics

- Exotic phases and phase transitions in equilibrium and non-equilibrium
- Symmetry, anomaly and anomaly matching

Mathematical Physics

- Algebraic topology, topological quantum field theory and its application in analyzing quantum phases

Machine Learning

- Classification of phases using machine learning methods

Quantum Information

- Applications of quantum error correcting code in condensed matter settings

PUBLICATION

Google Scholar Profile
Github Profile

- Arun Debray, **Weicheng Ye**, and Matthew Yu. "How to Build Anomalous (3+1)d Topological Quantum Field Theories." *arXiv preprint* (2025). [arXiv:2510.24834]
- **Weicheng Ye**, Shuwei Liu, Shiyu Zhou and Yijian Zou. "Universal quantum phase classification on quantum computers from machine learning." *arXiv preprint* (2025). [arXiv:2508.04774]
- Arun Debray, **Weicheng Ye**, and Matthew Yu. "Global structure in the presence of a topological defect." *arXiv preprint* (2025). [arXiv:2501.18399]
- Chunxiao Liu, and **Weicheng Ye**. "Crystallography, Group Cohomology, and Lieb-Schultz-Mattis Constraints." *SciPost Physics*, 18.1 (2025): 005. [arXiv:2410.03607]
- Rui Wen, **Weicheng Ye**, and Andrew C. Potter. "Topological Holography for fermions." *arXiv preprint* (2024). [arXiv:2404.19004]

- Arun Debray, **Weicheng Ye**, and Matthew Yu. "Bosonization and Anomaly Indicators of $(2+1)$-D Fermionic Topological Orders." *Commun. Math. Phys.*, 406, 178 (2025). [arXiv:2312.13341]
- Jinmin Yi, **Weicheng Ye**, Daniel Gottesman, and Zi-Wen Liu. "Complexity and order in approximate quantum error-correcting codes." *Nature Physics*, 20.11 (2024): 1798-1803. [arXiv:2310.04710]
- **Weicheng Ye**, and Liujun Zou. "Classification of symmetry-enriched topological quantum spin liquids." *Physical Review X*, 14.2 (2024): 021053. [arXiv:2309.15118]
- **Weicheng Ye**, and Liujun Zou. "Anomaly of $(2+1)$-Dimensional Symmetry-Enriched Topological Order from $(3+1)$-Dimensional Topological Quantum Field Theory." *SciPost Physics*, 15.1 (2023): 004. [arXiv:2210.02444]
- Cheng-Ju Lin, **Weicheng Ye**, Yijian Zou, Shengqi Sang, and Timothy H. Hsieh. "Probing sign structure using measurement-induced entanglement." *Quantum*, 7 (2023): 910. [arXiv: 2205.05692]
- **Weicheng Ye**, Meng Guo, Yin-Chen He, Chong Wang, and Liuju Zou. "Topological characterization of Lieb-Schultz-Mattis constraints and applications to symmetry-enriched quantum criticality." *SciPost Physics*, 13, no. 3 (2022): 066. [arXiv: 2111.12097]
- **Weicheng Ye**, Sung-Sik Lee, and Liujun Zou. "Ultraviolet-Infrared Mixing in Marginal Fermi Liquids." *Physical Review Letters*, 128, no. 10 (2022): 106402. [arXiv: 2109.00004][**Editor's Suggestion**]
- Bin Chen, Zhong-Ying Fan, Pengcheng Li, and **Weicheng Ye**. "Quasinormal modes of Gauss-Bonnet black holes at large D." *Journal of High Energy Physics*, 2016, no. 1 (2016): 1-27. [arXiv: 1511.08706]

PHD THESIS

Weicheng Ye. "Aspects of Anomaly in Condensed Matter Physics." (2023).

CONFERENCES AND TALKS

- Participants at the conference "Bootstrapping Nature: Non-perturbative Approaches to Critical Phenomena" in *The Galileo Galilei Institute For Theoretical Physics*.
- Talk at the seminar of *Institute for Advanced Study, Tsinghua University*. "Anomaly of Topological Order from TQFT"
- Invited poster at the conference "Quantum Entanglement and Topological Order" in *Chinese University of Hong Kong*. "Anomaly of Topological Order from TQFT"
- Talk at the seminar of *Fudan University*. "Anomaly of Topological Order: Symmetry Fractionalization, TQFT and Lattice Realization"
- Invited talk at the Postdoc Retreat in *University of Chinese Academy of Sciences*. "On the Classification and Construction of Topological Quantum Spin Liquid"
- Participant at the conference "Quantum Criticality and Topological Phases Transition" in *Chinese University of Hong Kong*.
- Invited talk at the workshop "Mini-Workshop on Unconventional Superconductivity and Correlated Electron Systems" in *Chinese University of Hong Kong*. "More Information from Lieb-Schultz-Mattis-Oshikawa-Hasting Constraints"
- Invited talk at the workshop "Applications of Generalized Symmetries and Topological Defects to Quantum Matter" in *Simons Center for Geometry and Physics*. "Lieb-Schultz-Mattis constraints and anomaly in 2d and 3d"

REFEREE EXPERIENCE

Referee for *Nature Communications, Physical Review Letter, PRX Quantum, Scipost Physics, Physical Review B, Physical Review A, Advances in Theoretical and Mathematical Physics, Journal of Function Spaces*

SKILLS

- Programming skills: Python, Julia, Mathematica, PyTorch
- Machine learning
- Mathematical modeling

OTHER EXPERIENCE

- PSI START Program Mentor *Perimeter Institute, 2022*
  Offering mentorship for upcoming graduate students on scientific projects
- Graduate Student Committee *Perimeter Institute, 2022*
  Furnish students with essential information, encompassing facility details, mental health resources, and other pertinent resources
- Reporter *Peking University, 2014*
  Deliver timely updates on significant scientific breakthroughs achieved within the physics department to undergraduate students

AWARDS

- **T. Alexander Pond Prize** *Stony Brook University, 2017*
- Zeng Xianzi Scholarship *Peking University, 2013 - 2015*
- **First Prize in Mathematical Contest of Modeling (MCM)** *Peking University, 2014*
- Weiming Scholarship for Meritorious Academic Achievements *Peking University, 2014/2015*
- **First Prize of Chinese Physics Olympiads (CPhO)**